

Sophia Robillard

Seattle, Washington | sophiaro@uw.edu | 206 588 3134 | sophiarobillard.github.io

Education

University of Washington - Seattle, WA Sept 2023 - Jun 2026

B.Sc. in Earth and Space Sciences: Geosciences, with Departmental Honors

- 3.54 GPA
- Minor: Data Science
- Dean's List: Winter 2026, Full Year 2025, Summer 2024, Winter 2024

University of California, Santa Barbara - Santa Barbara, CA Sept 2022 - Jun 2023

Research and Work Experience

Project Engineer - Machine Learning in the Geosciences, PaleoRocks Jan 2026 - Present

- Designed a convolutional neural network (CNN) to classify fossil and non-fossil features in petrographic thin section images, enhancing classification efficiency and enabling scalable analysis of geologic datasets in place of traditional point counting methods.
- Implemented Python-based/PyTorch machine learning workflows for image preprocessing, model training, and performance evaluation.

Undergraduate Researcher - Δ *IsoLab Sept 2024 - Present

Advised by Dr. Katharine Huntington - University of Washington

- Implement complex methods in carbonate clumped isotope geochemistry and petrography to investigate the origins of fine-grained lacustrine carbonate rocks and how micro-scale biological processes, microbially mediated carbonate formation conditions, and isotopic disequilibrium can alter geochemical data for paleoenvironment reconstructions.
- Awarded *Best Undergraduate Poster Presentation* at the 2025 Earth & Space Sciences (ESS) Gala

Environmental Geochemistry Student - Geomicrobiology Laboratory Mar 2024 - Jun 2024

- Assessed natural waters for contamination and quality parameters (pH, dissolved oxygen, conductivity, temperature), alkalinity, iron (II and III), dissolved organic carbon, and coliform bacteria.
- Performed alkalinity titrations, spectrophotometric analyses, and adsorption experiments relevant to low-temperature geochemistry.
- Gained experience in geochemical modeling softwares, PHREEQC and Equilibrium Diagrams, to model data found in environmental analyses. Reported findings, methods, and interpretations in cohesive laboratory reports.

Undergraduate Researcher - Isotope Geochemistry Laboratory Mar 2024 - Jun 2024

Advised by Dr. Bruce Nelson - University of Washington

- Learned techniques involving both purification of targeted elements through column chromatography in a clean sample preparation laboratory and high-precision isotopic analyses using MC-ICPMS.
- Collected preliminary $^{87}\text{Sr}/^{86}\text{Sr}$ isotope data from carbonate rock samples using mass-spectroscopy techniques.

Shift Lead & Event and Information Specialist - Husky Union Building Mar 2024 - Present

- Manage logistics for hundreds of high-profile meetings and events, providing immediate customer service and resolution of technical issues before and during events.
- Direct team members to execute complex event services and sets efficiently, ensuring client safety and satisfaction.

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Honors and Awards

- Earth & Space Sciences Departmental Honors Program 2024 - 2026
- Undergraduate Service Award - \$500 May 2026
- “The Money Figure” Best Figure ESS Gala - 2nd Place April 2026
- Afton Woolley Crooks & James William Crooks Endowed Scholarship April 2025
in Geological Sciences - \$13,407
- “*Rising Rockstar*” Best Undergraduate Poster Presentation ESS Gala - \$200 April 2025

Presentations

- **Robillard** et al., (2026). Formation Processes and Environments of Miocene Lacustrine Carbonates: Petrographic and Isotopic Implications for Paleoclimate Reconstruction on the Colorado Plateau, University of Washington Undergraduate Research Symposium, Seattle, WA. (Oral)
- **Robillard** et al., (2026). Formation Processes and Environments of Miocene Lacustrine Carbonates: Petrographic and Isotopic Implications for Paleoclimate Reconstruction on the Colorado Plateau, University of Washington Earth & Space Sciences Gala 2026, Seattle, WA. (Poster)
- **Robillard** et al., (2025). Petrographic and Isotopic Constraints on the Processes and Environments of Miocene Lacustrine Carbonate Formation on the Colorado Plateau, Geological Society of America Connects 2025, National Conference, San Antonio, TX. (Poster)
- **Robillard** et al., (2025). Decoding Lacustrine Carbonates from the Colorado Plateau: How Depositional Environments and Processes Shape the Stable Isotope Record, University of Washington Earth & Space Sciences Gala 2025, Seattle, WA. (Poster)

Forthcoming Publications

- **Robillard, S.**, Heitmann, E.O., Germeau, H., Leite, M., Schauer, A., Huntington, K.W. (in prep). Formation Processes and Environments of Miocene Lacustrine Carbonates: Petrographic and Isotopic Implications for Paleoclimate Reconstruction on the Colorado Plateau.
- Heitmann, E.O., Huntington, K.W., **Robillard, S.**, Fischer, L., Jorgensen, A., Mat, S., Leite, M., Brigham, C., He, J., Douglass, J., Cassel, E., Gootee, B., Schauer, A. (in prep). Lacustrine Carbonate Isotope Stratigraphy of the Miocene Bidahochi Formation Records Both Short-term and Long-term Change.

Service and Outreach

- Volunteer** - UW Rockin' Out Outreach K-12 Jan 2025 - Present
- Currently working with the graduate student leaders of the program to integrate and lead some new Rockin' Out initiatives I have designed. These include new activities, presentations, and potential new collaborations with Spiritridge Elementary in the Bellevue School District.
- Design and lead engaging Earth and space science lessons, incorporating hands-on demonstrations to inspire curiosity and foster a deeper understanding of geoscience concepts.
- Make complex topics accessible and exciting, encouraging students to ask questions, think critically, and explore their own scientific interests.

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Field Assistant - UW Department of Earth & Space Sciences

Sept 2025

- Measured 50+ meters of stratigraphic sections in the Wenatchee Formation and created stratigraphic columns from detailed notes and drawings done in the field.
- Collaborated with a small team to interpret the sedimentary rock record, collect samples for analysis, and record detailed outcrop observations.

Teaching Assistant (TAA) - Introduction to Geology (ESS 101)

Jan 2025 - Mar 2025

- Volunteered in weekly lab sections, aiding 20+ students' comprehension of geologic concepts and providing feedback on lab assignments. Met weekly with teaching staff to discuss lesson plans and lab session logistics.

After School Program Leader - Isla Vista Elementary School (K-6)

Oct 2022 - Jun 2023

- Organized educational enrichment activities for 30+ elementary students, including founding a "Science Club" to conduct hands-on physics and chemistry experiments.

Professional Skills

Software Proficiency

- Python
- ArcGISPro
- PHREEQC
- HTML & CSS
- Pytorch
- VSCode

Specialized Coursework

- Environmental Geochemistry
- Isotope Geochemistry
- Machine Learning
- Material Science
- Mechanics/Electromagnetism
- Sedimentary Geology

Languages

- English (Native)
- French (Certified B2, CEFR Scale)

Professional Affiliations

- Student Member of the Geological Society of America (GSA) - 2 Years